1. What is an algorithm?

An **algorithm** is a finite, step-by-step list of instructions, written in simple English, that solves a problem. It tells us what to do first, what to do next, and when to stop. Every step is numbered (Step 1, Step 2, ...) so that the order is clear.

2. Characteristics of a good algorithm

- **Input** - takes zero or more values (e.g. *Read n*).
- **Output** - gives at least one result (e.g. *Print sum*).
- **Definiteness** - every step is clear and has only one meaning.
- **Finiteness** - it must stop after a limited number of steps.
- **Effectiveness** - each step is simple enough to be carried out.

3. Rules for writing an algorithm

- Step 1 is always **Start** and the last step is always **Stop**.
- **Initialize** every variable (like i = 1, sum = 0, count = 0) before using it.
- Use **Read** to take input and **Print** to show output.
- Use **Check** for a decision. Write both exits: *if true goto step __, if false goto step __*.
- A **loop** is made with *goto*: after the last step of the loop body, goto the Check step again.

4. Keywords used

Keyword	Meaning	Example
Start / Stop	Beginning and end of the algorithm	Step 1: Start
Initialize	Give starting values to variables	Initialize i = 1, sum = 0
Read	Take input from the user	Read n
Print	Display an output	Print sum
Check	Test a condition (true / false)	Check i <= n
goto	Jump to another step (used for loops)	goto step 4

5. Operators used

Operator	Meaning	Example
=	Assign a value	sum = sum + i
==	Equal to (comparison)	n % i == 0
< <= > >=	Less / greater than (or equal)	i <= n
%	Remainder (modulus)	6 % 4 = 2, 253 % 10 = 3
/	Division (integer division for digits)	253 / 10 = 25
* ^	Multiply, power	f * i, d^c

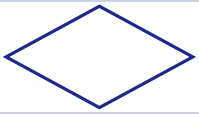
6. Example - algorithm to add two numbers

Step 1: Start **Step 2:** Read a, b **Step 3:** sum = a + b **Step 4:** Print sum **Step 5:** Stop

1. What is a flowchart?

A **flowchart** is the picture of an algorithm. It uses standard symbols joined by arrows to show the order in which the steps are carried out. Each step of the algorithm becomes one symbol.

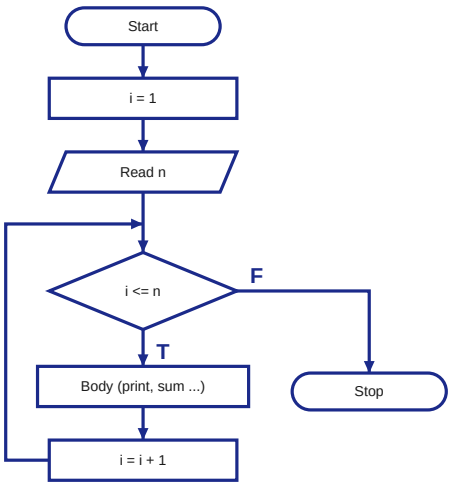
2. Flowchart symbols

Symbol	Name	Used for	Example
	Terminal (Oval)	Start and Stop of the flowchart	Start, Stop
	Input / Output (Parallelogram)	Reading input and printing output	Read n, Print i
	Process (Rectangle)	Calculations, assignments, initialization	sum = sum + i
	**Decision (Diamond)**	A condition with two exits: T (true) and F (false)	i <= n
	Flow line (Arrow)	Shows the direction of flow from one symbol to the next	
	Connector (Circle)	Joins two parts of a flowchart (or two pages)	A

3. Rules for drawing a flowchart

- Every flowchart has exactly one **Start** and one **Stop** oval.
- Draw from **top to bottom**; use arrows on every line.
- A **decision** diamond has one entry and two exits, labelled **T** (true) and **F** (false).
- Use a parallelogram for *Read* / *Print* and a rectangle for calculations.
- For a **loop**, draw an arrow from the end of the loop body back to the line just above the decision diamond.
- Keep symbols the same size and write short, clear text inside them.

4. How a loop looks in a flowchart



Initialize: sets i = 1 before the loop

Condition: i <= n is checked every time

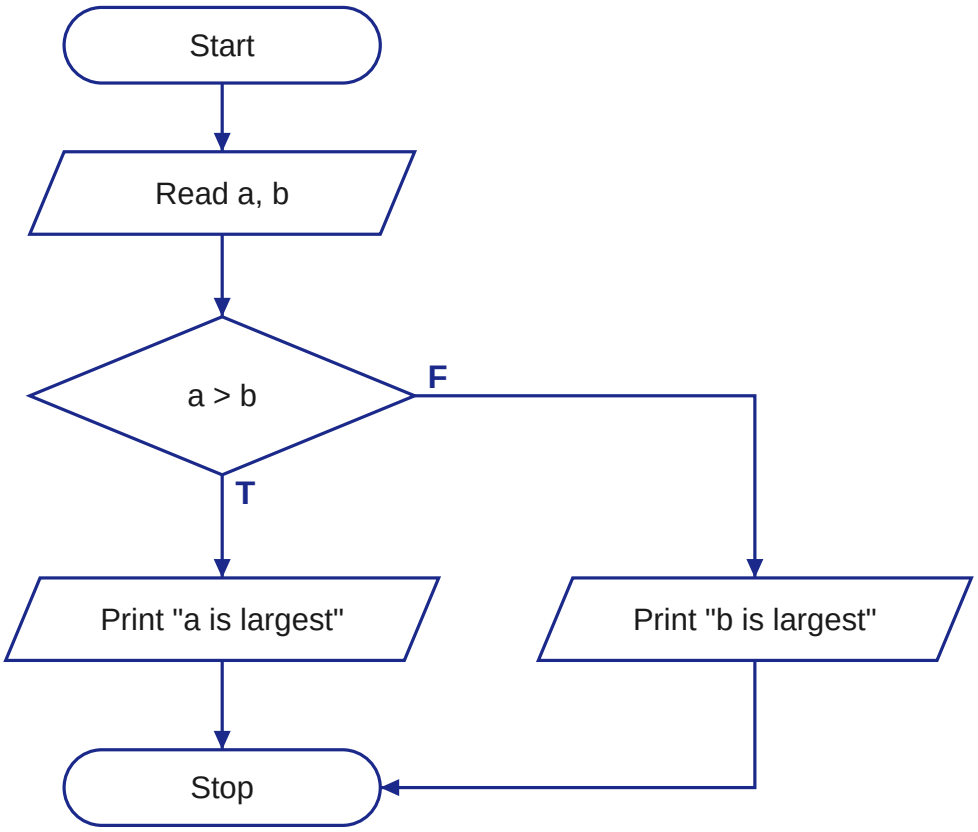
T path: body runs while condition is true

Update: i = i + 1 moves to the next value

Loop back: arrow returns above the diamond

F path: loop ends, goes to Stop / output

Q1. Largest of 2 numbers



Algorithm

Step 1: Start

Step 2: Read a, b

Step 3: Check a > b

if true goto step 4

if false goto step 5

Step 4: Print "a is largest", goto step 6

Step 5: Print "b is largest"

Step 6: Stop

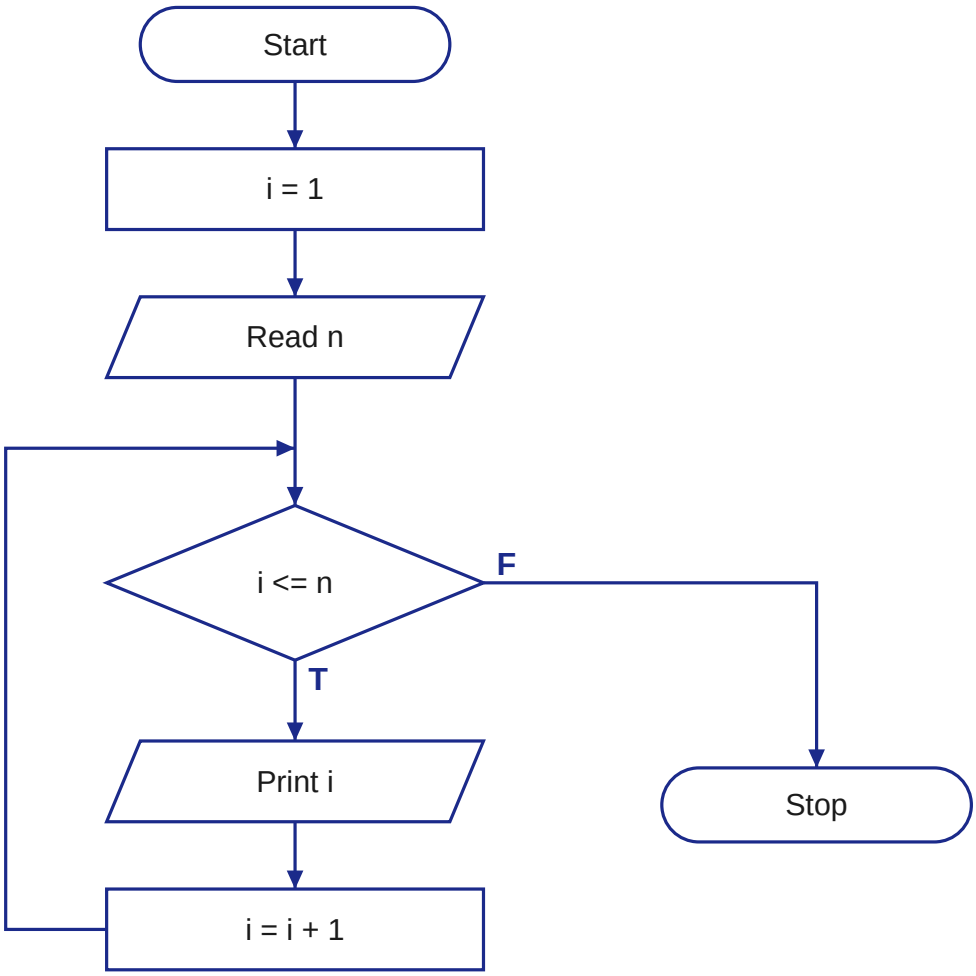
Example

Input: a = 5, b = 3

Output: a is largest

(If both are equal, either one is the largest.)

Q2. Print 1 to n



Algorithm

Step 1: Start

Step 2: Initialize i = 1

Step 3: Read n

Step 4: Check i <= n

if true goto step 5

if false goto step 7

Step 5: Print i

Step 6: i = i + 1, goto step 4

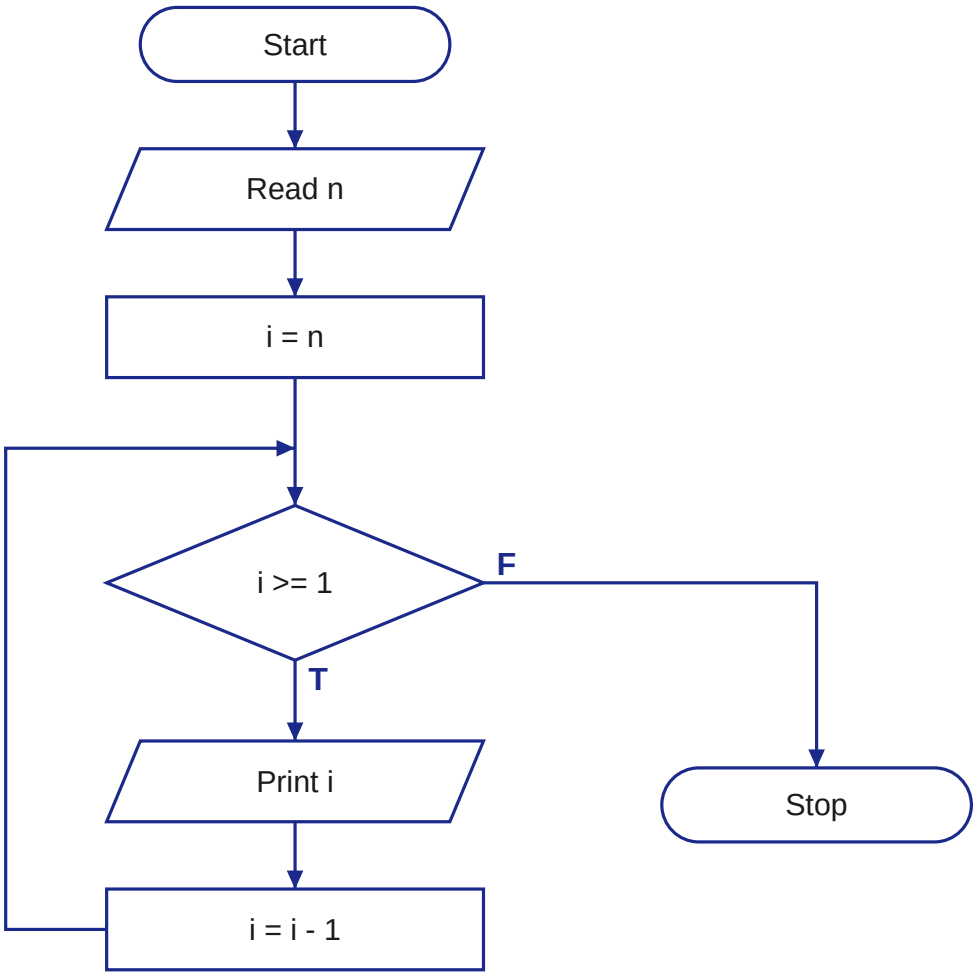
Step 7: Stop

Example

Input: n = 5

Output: 1 2 3 4 5

Q3. Print n to 1



Algorithm

Step 1: Start

Step 2: Read n

Step 3: Initialize i = n

Step 4: Check i >= 1

if true goto step 5

if false goto step 7

Step 5: Print i

Step 6: i = i - 1, goto step 4

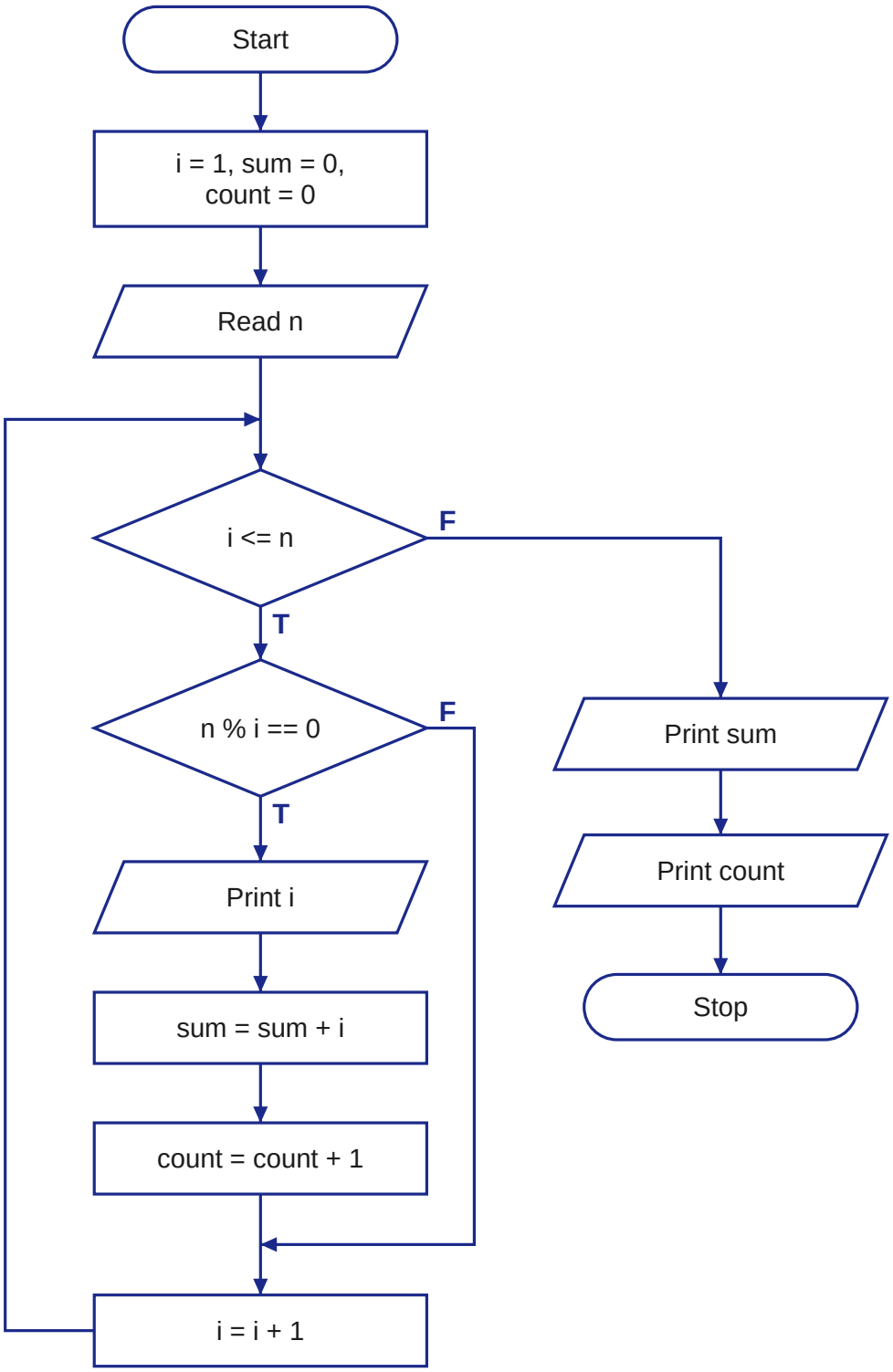
Step 7: Stop

Example

Input: n = 5

Output: 5 4 3 2 1

Q4. Print factors of a number, sum of factors, count



Algorithm

Step 1: Start

Step 2: Initialize i = 1, sum = 0, count = 0

Step 3: Read n

Step 4: Check i <= n

if true goto step 5

if false goto step 10

Step 5: Check n % i == 0

if true goto step 6

if false goto step 9

Step 6: Print i

Step 7: sum = sum + i

Step 8: count = count + 1

Step 9: i = i + 1, goto step 4

Step 10: Print sum

Step 11: Print count

Step 12: Stop

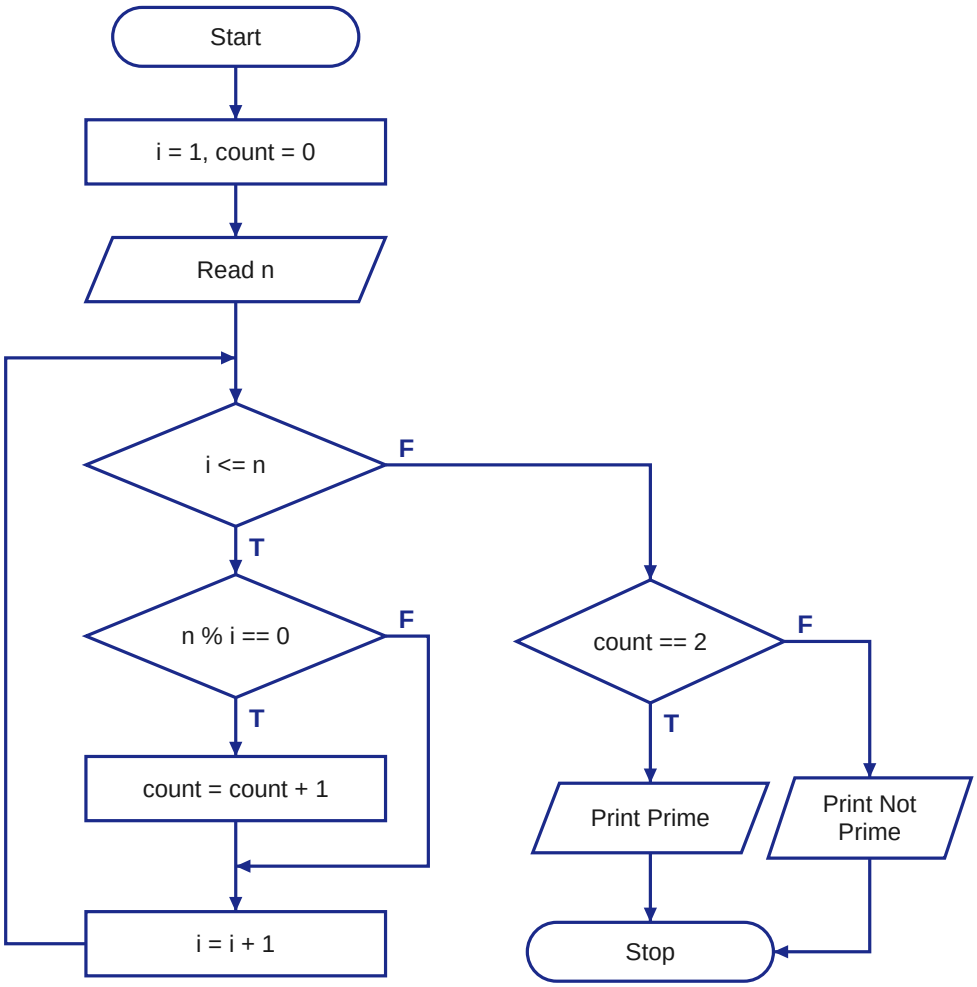
Example

Input: n = 6

Factors: 1 2 3 6

sum = 12, count = 4

Q5. Check prime or not



Algorithm

Step 1: Start

Step 2: Initialize i = 1, count = 0

Step 3: Read n

Step 4: Check i <= n

- if true goto step 5
- if false goto step 8

Step 5: Check n % i == 0

- if true goto step 6
- if false goto step 7

Step 6: count = count + 1

Step 7: i = i + 1, goto step 4

Step 8: Check count == 2

- if true goto step 9
- if false goto step 10

Step 9: Print "Prime", goto step 11

Step 10: Print "Not Prime"

Step 11: Stop

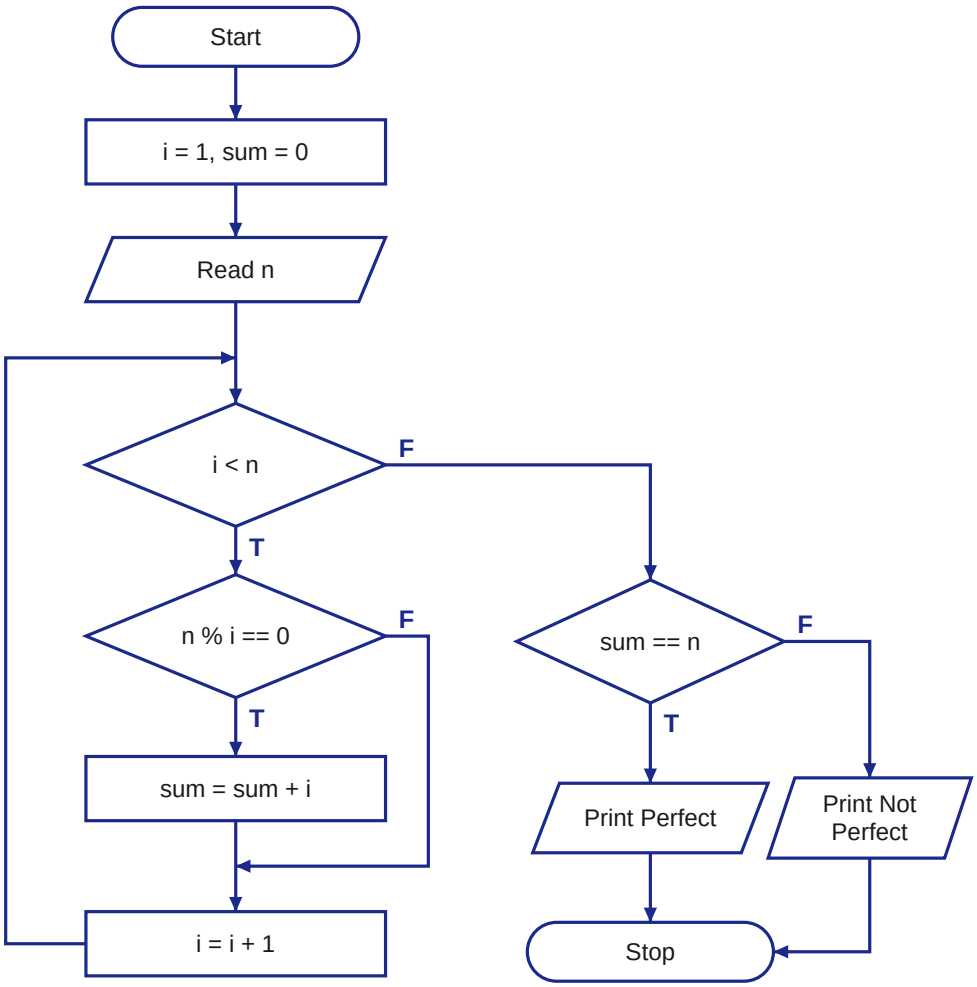
Example

Input: n = 7

Factors of 7 are 1 and 7, so count = 2

Output: Prime

Q6. Check perfect or not



Algorithm

Step 1: Start

Step 2: Initialize i = 1, sum = 0

Step 3: Read n

Step 4: Check i < n

if true goto step 5
if false goto step 8

Step 5: Check n % i == 0

if true goto step 6
if false goto step 7

Step 6: sum = sum + i

Step 7: i = i + 1, goto step 4

Step 8: Check sum == n

if true goto step 9
if false goto step 10

Step 9: Print "Perfect", goto step 11

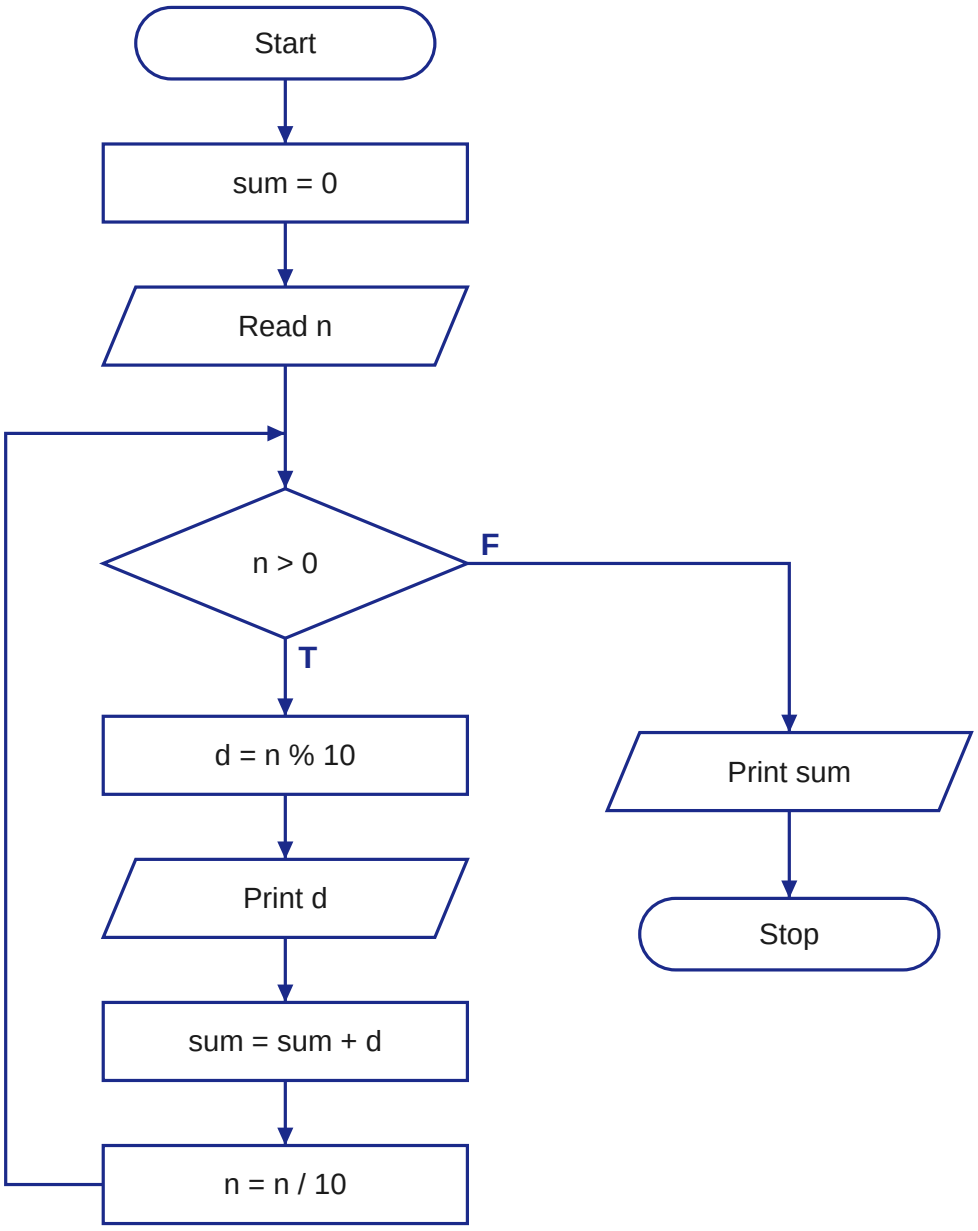
Step 10: Print "Not Perfect"

Step 11: Stop

Example

Input: n = 6
Factors less than 6: 1, 2, 3
sum = 1+2+3 = 6, Output: Perfect

Q7. Digits of a number, sum of digits



Algorithm

Step 1: Start

Step 2: Initialize sum = 0

Step 3: Read n

Step 4: Check n > 0

if true goto step 5

if false goto step 9

Step 5: d = n % 10

Step 6: Print d

Step 7: sum = sum + d

Step 8: n = n / 10, goto step 4

Step 9: Print sum

Step 10: Stop

Example

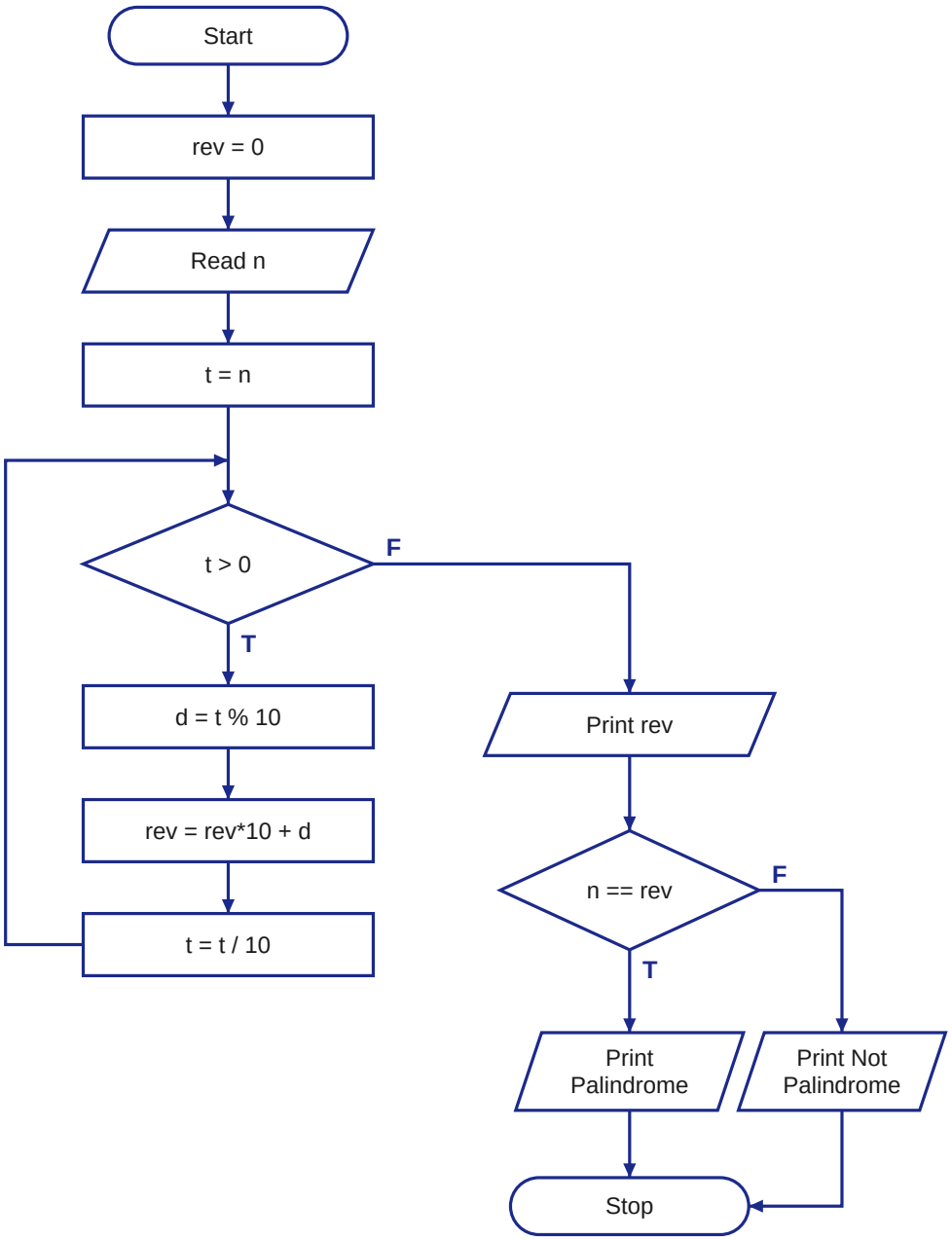
Input: n = 253

Digits printed: 3 5 2 (last digit first)

Sum of digits = 10

(n / 10 means integer division: 253 / 10 = 25)

Q8. Print reverse and check palindrome or not



Algorithm

Step 1: Start

Step 2: Initialize rev = 0

Step 3: Read n

Step 4: t = n

Step 5: Check t > 0

if true goto step 6

if false goto step 9

Step 6: d = t % 10

Step 7: rev = rev * 10 + d

Step 8: t = t / 10, goto step 5

Step 9: Print rev

Step 10: Check n == rev

if true goto step 11

if false goto step 12

Step 11: Print "Palindrome", goto step 13

Step 12: Print "Not Palindrome"

Step 13: Stop

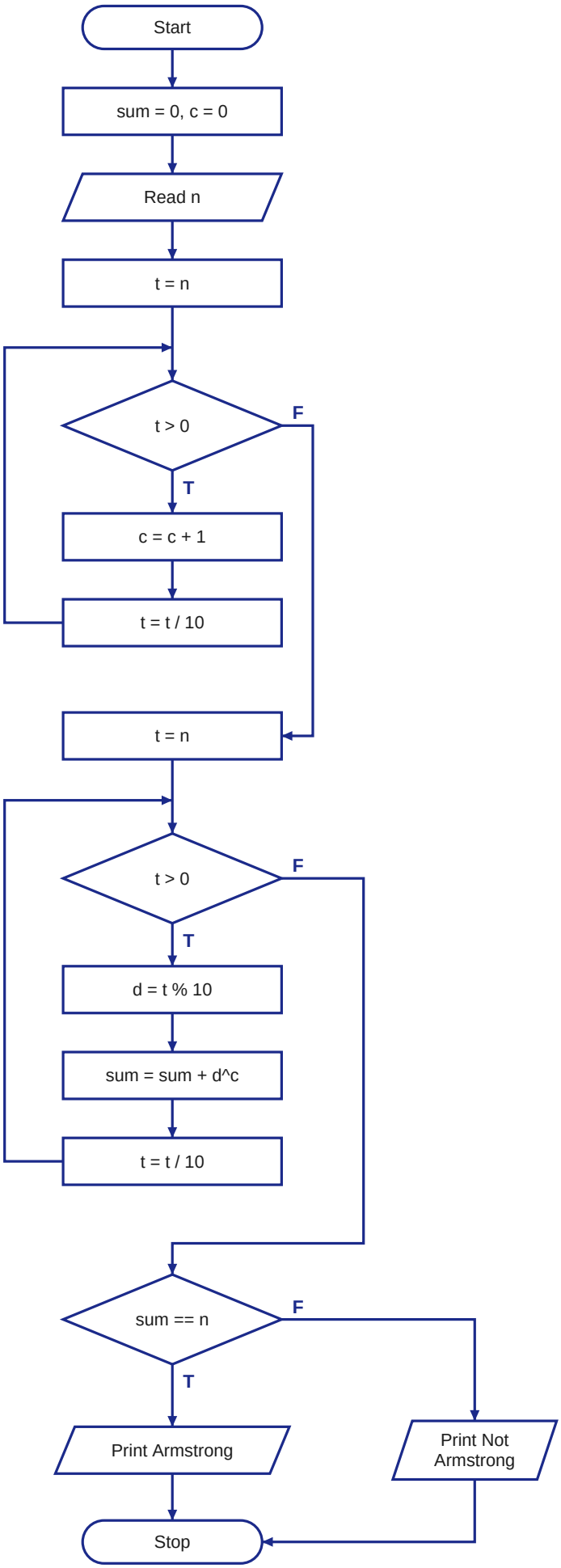
Example

Input: n = 121

rev = 121, so n == rev

Output: 121, Palindrome

Q9. Check Armstrong or not



Algorithm

Step 1: Start

Step 2: Initialize sum = 0, c = 0

Step 3: Read n

Step 4: t = n

Step 5: Check t > 0

if true goto step 6

if false goto step 8

Step 6: c = c + 1 (counts digits)

Step 7: t = t / 10, goto step 5

Step 8: t = n

Step 9: Check t > 0

if true goto step 10

if false goto step 13

Step 10: d = t % 10

Step 11: sum = sum + d^c

Step 12: t = t / 10, goto step 9

Step 13: Check sum == n

if true goto step 14

if false goto step 15

Step 14: Print "Armstrong", goto step 16

Step 15: Print "Not Armstrong"

Step 16: Stop

Example

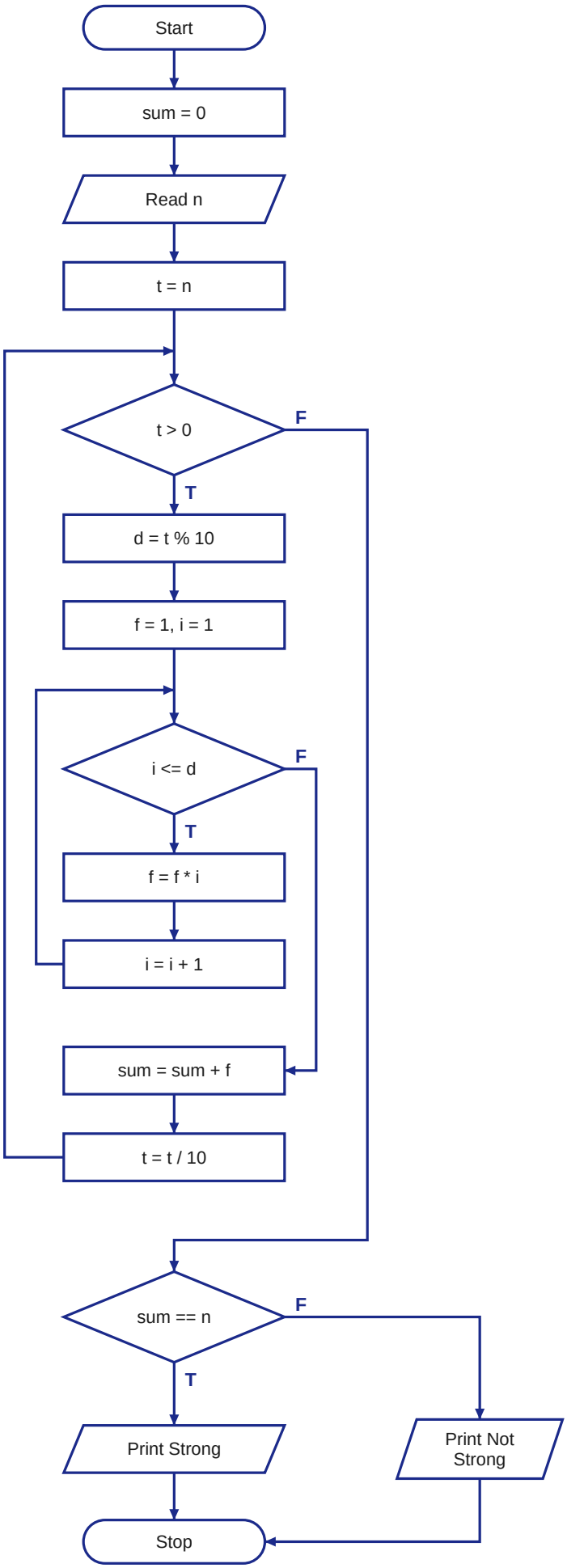
Input: n = 153, so c = 3 digits

1^3 + 5^3 + 3^3 = 1+125+27 = 153

Output: Armstrong

(d^c means d raised to the power c)

Q10. Check strong or not



Algorithm

Step 1: Start

Step 2: Initialize sum = 0

Step 3: Read n

Step 4: t = n

Step 5: Check t > 0

if true goto step 6

if false goto step 13

Step 6: d = t % 10

Step 7: f = 1, i = 1

Step 8: Check i <= d

if true goto step 9

if false goto step 11

Step 9: f = f * i

Step 10: i = i + 1, goto step 8

Step 11: sum = sum + f

Step 12: t = t / 10, goto step 5

Step 13: Check sum == n

if true goto step 14

if false goto step 15

Step 14: Print "Strong", goto step 16

Step 15: Print "Not Strong"

Step 16: Stop

Example

Input: n = 145

1! + 4! + 5! = 1+24+120 = 145

Output: Strong

(f is the factorial of each digit d)